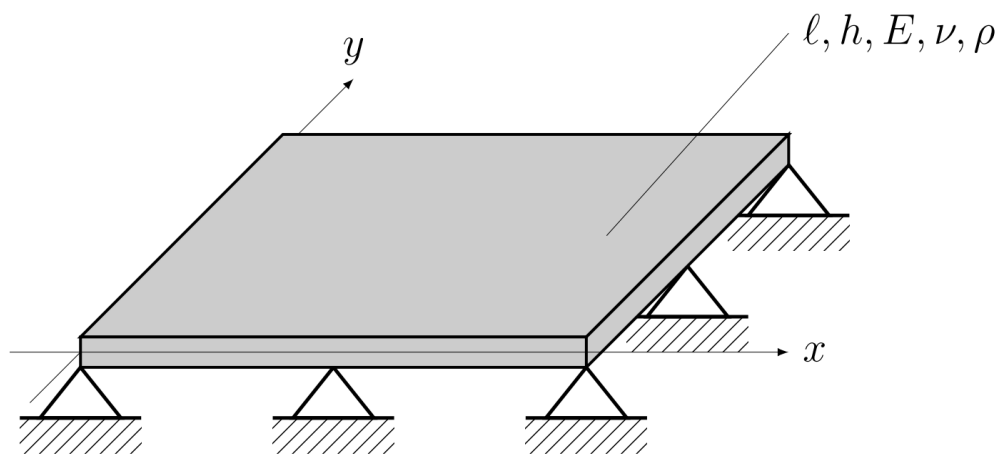




ME-473
MINI-PROJECT 3

Natural frequencies and mode shapes of a square plate



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1 Introduction

This project investigates the accuracy of various thin-plate methods in estimating the first natural frequency of a square, isotropic, thin plate that is simply supported along all four edges. The number of elements, the thickness of the plate, and the type of elements used is varied, and the first natural frequency is compared to thin and thick plate analytic estimations. The symmetry of the plate allows for analysis to be performed on a quarter of the plate to reduce calculation time.

2 Methodology

2.1 Material and geometrical properties

The geometric and material quantities of the plate used in the analysis are as follows:

- $L = 0.5 [m]$: length of the edge
- $h = 0.005 [m]$: thickness of the plate
- $E = 210 [GPa]$: Young's modulus of steel
- $\nu = 0.3 [-]$: Poisson's ratio
- $\rho = 7850 [kg/m^3]$: mass density of steel

2.2 Governing equation and boundary conditions

The strong form governing the isotropic plate is defined as

$$\begin{cases} D \nabla_k^T \mathbf{C}^* \nabla_k u_3 + \rho h \ddot{u} = 0 & \Omega \times [0, T] \\ u_3 = 0 & \partial\Omega \\ M = 0 & \partial\Omega \end{cases} \quad (1)$$

where D is the flexural rigidity of the material, the constitutive matrix $\mathbf{C}^*$ and ∇_k is the curvature gradient operator. These quantities are defined as follow.

$$D = \frac{Eh^3}{12(1-\nu^2)}, \quad \mathbf{C}^* = \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}, \quad \nabla_k^2 = \begin{bmatrix} \frac{\partial^4}{\partial x_1^4} & \frac{\partial^4}{\partial x_2^4} & 2 \frac{\partial^4}{\partial x_1^2 \partial x_2^2} \end{bmatrix}^T \quad (2)$$

Multiplying by the virtual displacement and integrating by parts twice while applying the boundary conditions, gives the weak form of the problem with respective function spaces.

$$\int_{\Omega} D \nabla_k u_3 \mathbf{C}^* \nabla_k \delta u_3 d\Omega + \int_{\Omega} \rho h \ddot{u} \delta u d\Omega = 0 \quad (3)$$

$\mathbf{K}$ to obtain the reduced matrices $\mathbf{M}_{\text{red}}$ and $\mathbf{K}_{\text{red}}$. In order to find the natural frequencies and mode shapes, the weak form is turned into an eigenvalue problem.

$$\mathbf{K}_{\text{red}} {}^e\mathbf{q} + \mathbf{M}_{\text{red}} {}^e\ddot{\mathbf{q}} = 0 \quad \Rightarrow \quad (\mathbf{K}_{\text{red}} - 4\pi f^2 \mathbf{M}_{\text{red}}) \boldsymbol{\phi} = 0 \quad (12)$$

where f are the natural frequencies in Hz each associated to a modal vector $\boldsymbol{\phi}$.

2.3 Adini-Clough-Melosh (ACM) method

This method prescribes three degrees of freedom (DOF) per node. They are, in order, u_3 the vertical displacement followed by θ_1 and θ_2 the bending slopes about x_1 and x_2 respectively. Inspecting a single rectangular element, we thus obtain a total of $4 \times 3 = 12$ total dofs per element. This means that at least twelve independent displacement functions must be selected (not to be confused with shape functions). The set of displacement functions should also include a cubic polynomial for each direction x_1 and x_2 as they allow the plate element to have a non-constant curvature. Indeed, a cubic polynomial becomes linear when differentiated twice.

$$\mathcal{P}_\phi^{12} = \{1, x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, x_1x_2^2, x_2^3, x_1^3x_2, x_1x_2^3\}$$

This set allows the approximated vertical displacement u_3 to be expressed as the sum of each displacement function ϕ_i .

$${}^e u_3^h(x_1, x_2, t) = \sum_{i=1}^{12} \phi_i(x_1, x_2) \alpha_i(t) = \boldsymbol{\Phi} \boldsymbol{\alpha}(t) \quad (13)$$

We can also rewrite this expression as the multiplication of the shape function matrix ${}^a H(\xi)$ and the nodal dof vector ${}^e q(t)$.

$${}^e u_3^h(x_1, x_2, t) = \sum_{i=1}^4 {}^a h_i(\xi) {}^e q(t) = {}^a H(\xi) {}^e q(t) \quad (14)$$

where ${}^a H(\xi) = \begin{bmatrix} {}^a h_1(\xi) & {}^a h_2(\xi) & {}^a h_3(\xi) & {}^a h_4(\xi) \end{bmatrix}$ and ${}^e q(t) = \begin{bmatrix} {}^a q_1(t) & {}^a q_2(t) & {}^a q_3(t) & {}^a q_4(t) \end{bmatrix}^T$.

As discussed previously, each one of the four nodes of the archetypal rectangle contains three DOFs grouped inside ${}^a q_i(t)$. The former is defined more formally as

$${}^e q_i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\xi^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\xi^i, t)/b \\ -\partial_{\xi_1} {}^e u_3^h(\xi^i, t)/a \end{bmatrix} \quad (15)$$

where the a and b factors come from the coordination transformation $\xi \mapsto x(\xi)$ and more specifically $dx_1 = a d\xi_1$ and $dx_2 = b d\xi_2$.

In order to link the displacement functions ϕ_i with the shape functions $^a h_i$, Eq.13 and 15 are combined to rewrite $^e q^i(t)$ for any node i .

$$^e q^i(t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & -1/a \end{bmatrix} \begin{bmatrix} 1 \\ \partial_{\xi_1} \\ \partial_{\xi_2} \end{bmatrix} \Phi \alpha(t) = \bar{P}(\xi^i) \alpha(t) \rightarrow \bar{P}(\xi^i) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & -1/a \end{bmatrix} P(\xi^i) \quad (16)$$

$$P(\xi^i) = \begin{bmatrix} 1 & \xi_1 & \xi_2 & \xi_1^2 & \xi_1 \xi_2 & \xi_2^2 & \xi_1^3 & \xi_1^2 \xi_2 & \xi_1 \xi_2^2 & \xi_2^3 & \xi_1^3 \xi_2 & \xi_1 \xi_2^3 \\ 0 & 0 & 1 & 0 & \xi_1 & 2\xi_2 & 0 & \xi_1^2 & 2\xi_1 \xi_2 & 3\xi_2^2 & \xi_1^3 & 3\xi_1 \xi_2^2 \\ 0 & 1 & 0 & 2\xi_1 & \xi_2 & 0 & 3\xi_1^2 & 2\xi_1 \xi_2 & \xi_2^2 & 0 & 3\xi_1^2 \xi_2 & \xi_2^3 \end{bmatrix}$$

where P and $\bar{P}$ represent a singular node. A is now defined as the 12×12 matrix composed of the different $\bar{P}(\xi^i)$ over the four nodes. Using the fact that $^e q(t) = A \alpha(t)$, this means that we can write $\alpha(t) = A^{-1} ^e q(t)$ and injecting it back into Eq. 13 gives us:

$$\begin{bmatrix} ^e q^1(t) \\ ^e q^2(t) \\ ^e q^3(t) \\ ^e q^4(t) \end{bmatrix} = \begin{bmatrix} \bar{P}(\xi^1) \\ \bar{P}(\xi^2) \\ \bar{P}(\xi^3) \\ \bar{P}(\xi^4) \end{bmatrix} \alpha(t) = A \alpha(t) \quad (17)$$

$$^e u_3^h(x, y, t) = \Phi \alpha(t) = \Phi \begin{bmatrix} \bar{P}(\xi^1) \\ \bar{P}(\xi^2) \\ \bar{P}(\xi^3) \\ \bar{P}(\xi^4) \end{bmatrix}^{-1} \begin{bmatrix} ^e q^1(t) \\ ^e q^2(t) \\ ^e q^3(t) \\ ^e q^4(t) \end{bmatrix} = \Phi A^{-1} ^e q(t) \quad (18)$$

We can verify if the dimensions are coherent.

$$^e u_3^h : \mathbb{R}^{1 \times 1}, \quad \Phi : \mathbb{R}^{1 \times 12}, \quad A^{-1} : \mathbb{R}^{12 \times 12}, \quad ^e q(t) : \mathbb{R}^{12 \times 1}$$

$$1 \times 1 \stackrel{?}{=} (1 \times 12) \cdot (12 \times 12) \cdot (12 \times 1) = (1 \times 12) \cdot (12 \times 1) = 1 \times 1 \quad \checkmark$$

After plugging-in that result in Eq. 14, we finally can compute $^a H(\xi)$ as

$$^a H(\xi) ^e q(t) = \Phi A^{-1} ^e q(t) \Rightarrow ^a H(\xi) = \Phi A^{-1} \quad (19)$$

where $^a H(\xi)$ is 1×12 and being composed of $^a h_i(\xi)$, with $\xi^i = \{\xi_1^i, \xi_2^i\}$ the local coordinate of node i , defined as

$$^a h_i(\xi) = \begin{bmatrix} (1 + \xi_1^i \xi_1) (1 + \xi_2^i \xi_2) (2 + \xi_1^i \xi_1 + \xi_2^i \xi_2 - \xi_1^2 - \xi_2^2) / 8 \\ b (1 + \xi_1^i \xi_1) (\xi_2^i + \xi_2) (\xi_2^2 - 1) / 8 \\ -a (1 + \xi_2^i \xi_2) (\xi_1^i + \xi_1) (\xi_1^2 - 1) / 8 \end{bmatrix}^T \quad (20)$$

Once the shape function matrix $^a H$ is obtained, the element mass matrix $^e M$ and element stiffness matrix $^e K$ can be computed using Eq. 8 and 11.

2.4 Crouzeix-Raviart (CR) method

The Crouzeix-Raviart method follows a similar strategy for deducing shape functions, but this time using rectangular elements with four nodes each having four degrees of freedom per node. These degrees of freedom are the vertical displacement u_3 , the bending slopes about x_1 and x_2 called θ_1 and θ_2 , and finally the twist slope θ_{12} . Therefore, each node of the archetypal element has four DOFs defined as:

$${}^e q_i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \\ {}^e \theta_{12}^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\xi^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\xi^i, t)/b \\ -\partial_{\xi_1} {}^e u_3^h(\xi^i, t)/a \\ -\partial_{\xi_1 \xi_2}^2 {}^e u_3^h(\xi^i, t)/(ab) \end{bmatrix} \quad (21)$$

This additional degree of freedom is added so that the plate no longer tends to a zero twist condition when the number of elements used is increased. The displacement approximation for these elements contains a full cubic polynomial as well as quartic, quintic and one sextic term (ACM elements only added quartic terms to the cubic polynomial):

$$\mathcal{P}_\phi^{16} = \{1, x_1, x_2, x_1^2, x_1 x_2, x_2^2, x_1^3, x_1^2 x_2, x_1 x_2^2, x_2^3, x_1^3 x_2, x_1 x_2^3, x_1^2 y^2, x^3 y^2, x^2 y^3, x^3 y^3\}$$

The approximate vertical displacement may be defined as the combination of shape functions as follows:

$${}^e u_3^h(x_1, x_2, t) = \sum_{i=1}^4 {}^a h_i(\xi) {}^e q_i(t) = {}^a H(\xi) {}^e q(t)$$

where ${}^a H(\xi) = \begin{bmatrix} {}^a h_1 & {}^a h_2 & {}^a h_3 & {}^a h_4 \end{bmatrix}$. The same process as was followed with the ACM method is used to link the displacement functions with the shape functions, followed by creating the element mass matrix and element stiffness matrix (Eq. 8-11). As a result of this process, the shape functions are defined as follows:

$${}^a h_i(\xi) = \begin{bmatrix} f_i(\xi_1) f_i(\xi_2) \\ b f_i(\xi_1) g_i(\xi_2) \\ -a g_i(\xi_1) f_i(\xi_2) \\ a b g_i(\xi_1) g_i(\xi_2) \end{bmatrix}^T \quad (22)$$

where a and b indicate half the length and half the width of the element, respectively. The functions f_i and g_i are defined as:

$$f_i(\xi) = (-\xi^i \xi^3 + 3\xi^i \xi + 2)/4$$

$$g_i(\xi) = (\xi^3 + \xi^i \xi^2 - \xi - \xi^i)/4$$

where ξ^i indicates the coordinate of node i .

3 Results and discussion

The impact of element type, number of elements, and plate length to height ratio on the accuracy of the calculated first natural frequency is investigated using ACM and CR elements, with 2×2 to 16×16 span-wise elements and length to height ratios from 10 to 100. All cases tested and their results are shown in Table 1. Percent error is found by comparing the first natural frequency calculated using thin-plate elements to two different analytical models. Firstly, the Kirchhoff-Love (KL) model:

$$f_{m,n}^{KL} = \frac{\pi}{2} \frac{m^2 + n^2}{L^2} \sqrt{\frac{D}{\rho h}} \quad (23)$$

Secondly, the Reissner-Mindlin (RM) model:

$$f_{m,n}^{RM} = F(L, h, E, \nu, \rho, m, n) \quad (24)$$

where this function cannot be expressed in a simple compact form. For thicker plate elements, the Reissner-Mindlin method is better adapted as it takes into account the surface shear, while Kirchhoff-Love does not.

Element Type	# of Elements	l/h ratio	f_{n1}	Error (%) KL	Error (%) RM
ACM	2	100	89.7840	8.7032	8.6774
ACM	4	100	95.4066	2.9858	2.9585
ACM	8	100	97.5447	0.8117	0.7838
ACM	16	100	98.1391	0.2073	0.1792
CR	2	100	98.5550	0.2156	0.2438
CR	4	100	98.3561	0.0133	0.0415
CR	8	100	98.3438	0.0008	0.0290
CR	16	100	98.3430	0.0001	0.0282
ACM	2	10	897.8402	8.7032	6.1640
ACM	4	10	954.0663	2.9858	0.2877
ACM	8	10	975.4470	0.8117	1.9469
ACM	16	10	981.3908	0.2073	2.5681
CR	2	10	985.5499	0.2156	3.0028
CR	4	10	983.5610	0.0133	2.7949
CR	8	10	983.4380	0.0008	2.7821
CR	16	10	983.4303	0.0001	2.7813

Table 1: First Natural Frequency results for various element types, number of elements, and length to thickness ratios. Here, KL refers to the Kirchhoff-Love analytical model, and RM refers to the Reissner-Mindlin analytical model.

The exact first mode shape for thin plates is given by:

$$u_3(x, y, t) = \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) (A \sin(2\pi f_1 t) + B \cos(2\pi f_1 t)) \quad (25)$$

where A and B depend on the initial conditions.

3.1 ACM vs CR Elements

CR elements show much more rapid convergence towards the Kirchoff-Love analytical solution than ACM elements (Figure 1). When only 2 span-wise elements are used, the CR element estimation has an error 97.2% less than that achieved with ACM elements (Table 1)! In order to maintain an error of less than 1%, 8 span-wise elements are needed with the ACM method while only 2 are needed for the CR method. Indeed, the CR method converges exponentially with a power of 4 while the ACM method has much slower convergence, with a power of 2 (Figure 2).

Additionally, this rapid convergence behavior does not require significantly more computation time. For the 16 element case, the ACM element estimation required 0.41 seconds to compute, while the CR element estimation required 0.57 seconds (both using Gauss-Legendre integral approximations and taking advantage of the plate symmetry so as to only model 1/4 of the plate). 0.1 seconds out of the 0.16 second difference comes from solving the eigenvalue problem which takes twice as long for the CR elements compared to the AMCs; the rest of the difference comes from the assembly of the rigidity matrix.

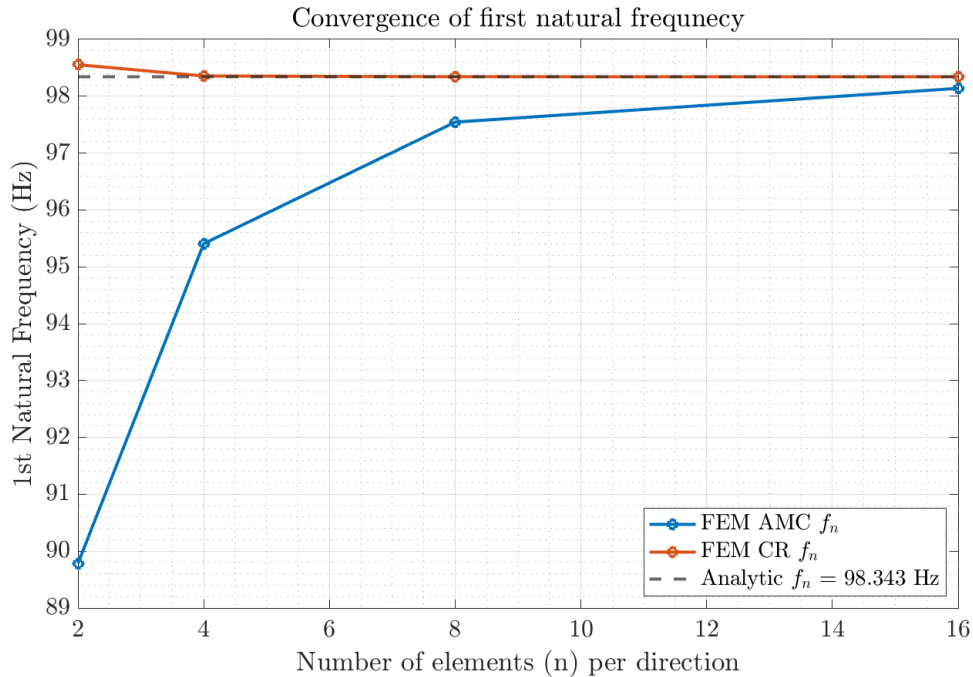


Figure 1: Convergence of the first natural frequency with respect to the number of elements with the two methods. Here, $n = 4$ corresponds to a mesh of size 4×4 elements in the global domain.

Of note, the CR method overestimates the first natural frequency, while the ACM method underestimates it. The ACM method uses non-conforming shape functions that do not satisfy full C^1 continuity across element boundaries, particularly in rotations. C^1 continuity means that a function and its first derivative is continuous, and this is a continuity required by the Kirchhoff plate theory. The non-conformity to the C^1 condition creates reduced inter-element bending stiffness, leading to a softer stiffness matrix and allowing more deformation than physically accurate. This creates an underestimation of the natural frequency. Thus, this model is mostly suitable for cases when small slope discontinuities are acceptable. In contrast, the CR method tends to produce a stiffer global stiffness matrix than the ACM method. Its extra DOF at each node helps with convergence and ensures full conformity to C^1 continuity, but augments the constraints on the displacement field. This additional stiffness restricts the displacement field, particularly for the first (lowest) mode, resulting in an overestimation of the first natural frequency.

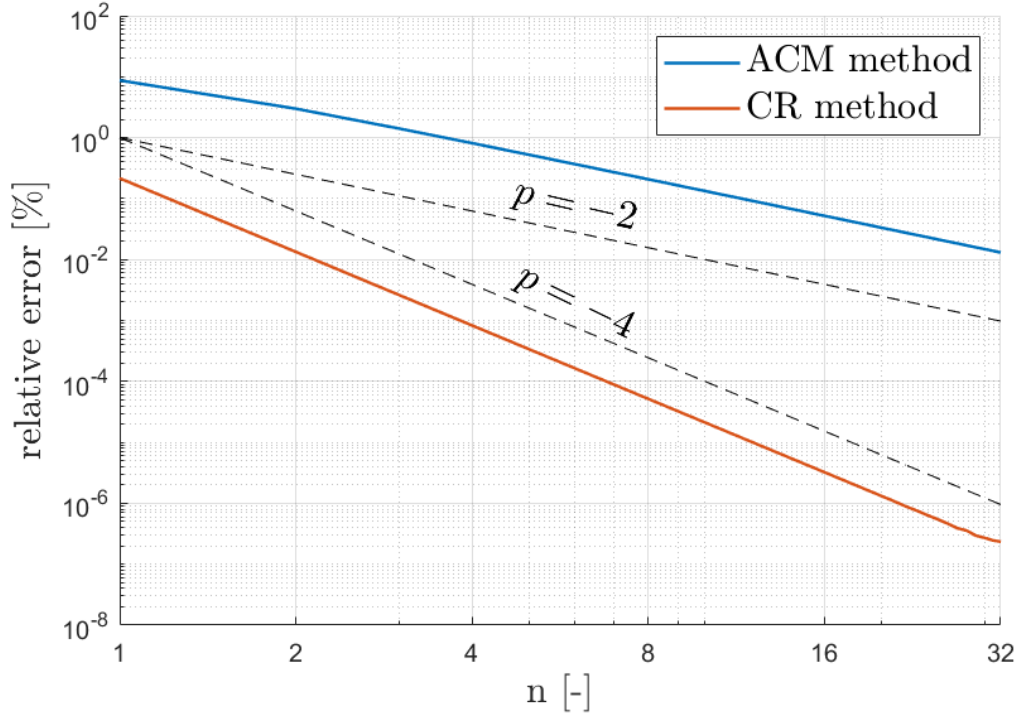


Figure 2: Convergence of the first natural frequency in log-space. The black lines correspond to functions defined as $f(n) = n^p$. Here, $n = 2$ corresponds to a mesh of size 4×4 elements in the global domain, as only a quarter of the plate was used to perform analysis.

Since the CR method uses conformal elements, it is possible to derive an upper bound for the error estimation as a function of the element size. The following error estimate boundaries are defined as (Reference 1):

$$\omega_i \leq \omega_i^h \leq \omega_i + \bar{c}h^{2(m-k+1)}\omega_i^{2m+1} \quad (26)$$

- ω_i and ω_i^h are the exact and approximated frequencies, respectively

- h represents the characteristic mesh size
- m is the degree of the highest complete polynomial used
- $\bar{c}$ is a constant independent of h
- k denotes the highest derivative order appearing in the weak form

For CR elements, the maximum degree of the polynomial used is 3 for both ξ_1 and ξ_2 , meaning that m is equal to 3. The highest order of derivatives, k , in the weak form is 2. Then using Eq.26, the relative error is bounded from above by the function $\bar{c}h^4$ which aligns with the observation obtained when using the finite element methods.

3.2 Mode Shape

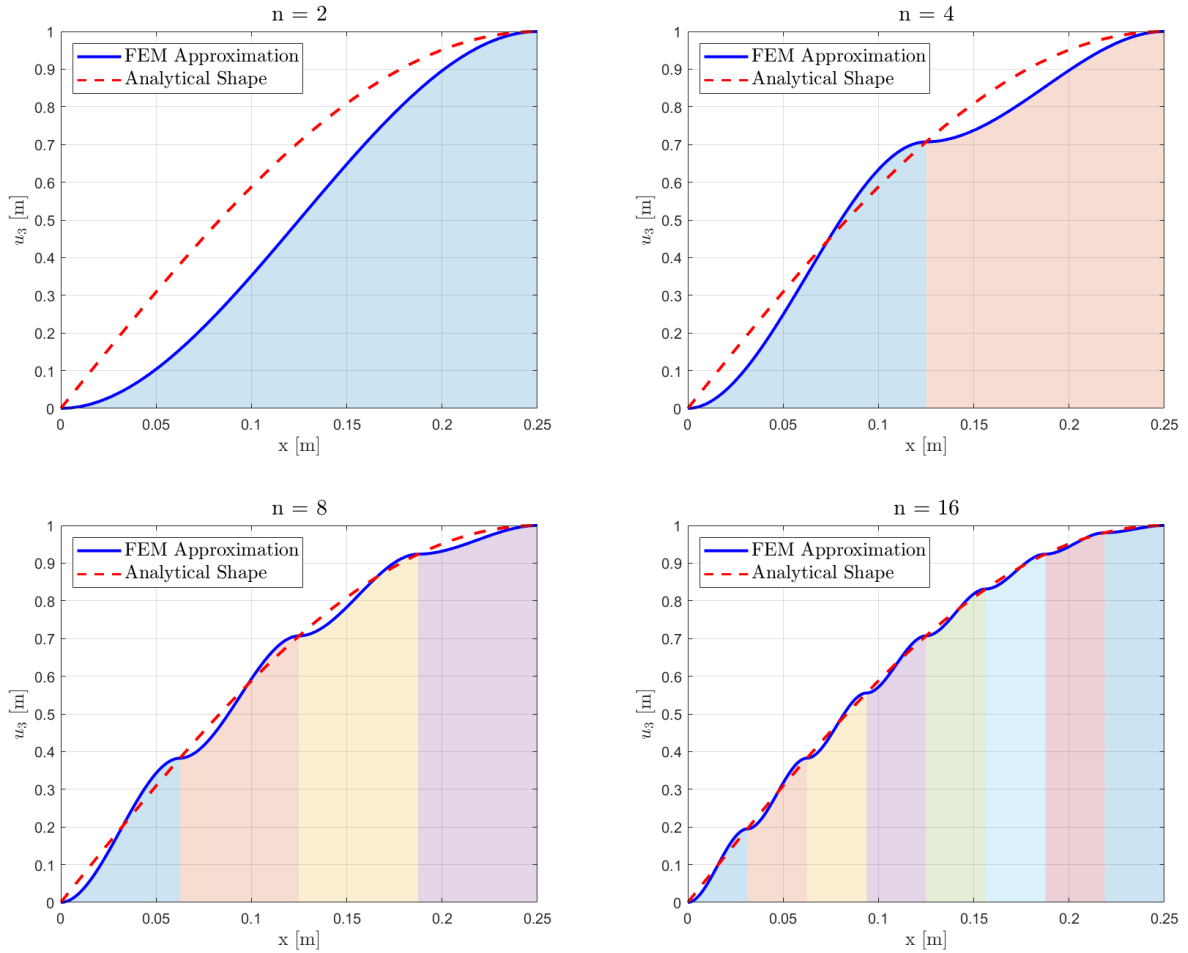


Figure 3: Approximated first mode shape compared to the analytical shape over half of the midline of the plate for different mesh sizes. The ACM and CR methods give the exact same results. The FEM approximations were computed using the corresponding shape functions along the full x direction instead of taking only the displacements at the nodal locations. Each color represents one element of the mesh.

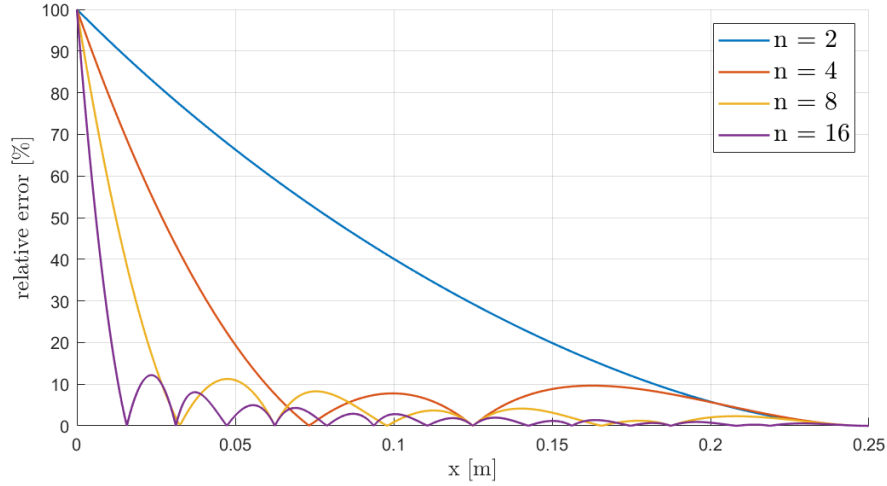


Figure 4: Relative error between the FEM mode shape approximation and the analytical shape (Equation 25).

As observed in Fig.3, the nodal displacements surprisingly match the first mode shape while the intermediary points follow a relative error seen in Fig.4. Despite using only piecewise cubic polynomials, the projection of the first mode shape onto the finite element space coincides exactly at the nodes. More precisely, the first eigenvector obtained from modal analysis and approximating the first mode shape is identical to the uniformly discretized first mode shape. This happens often for symmetric problems with smooth solutions which align well with the mesh and with the shape functions. This property is called *superconvergence*.

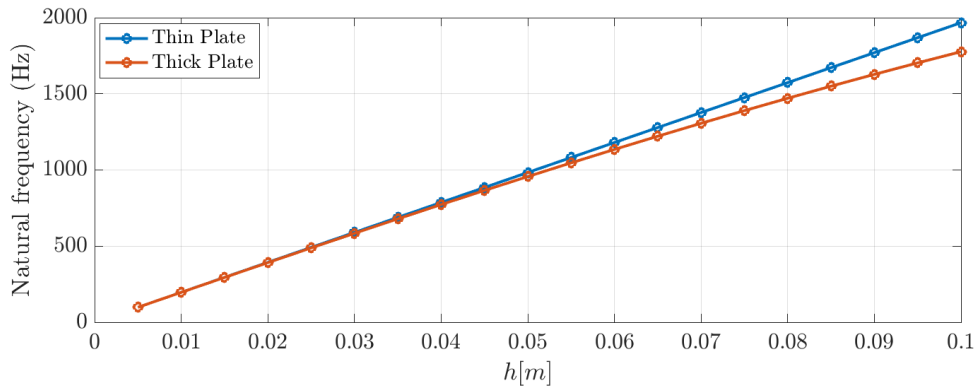
On another note, the derivatives in any direction of the shape functions associated with u_3 are zero along the edges of the element. This is expected, since in Kirchhoff-Love theory bending is governed by the first derivatives of the transverse displacement u_3 . The bending behavior is represented by two separate rotational degrees of freedom at each node. Therefore, the shape functions associated solely with u_3 must have zero first derivatives along the edges, so as not to interfere with the rotational (bending) DOFs. This characteristic is most visible for $n = 2$ where in this case the FEM approximation is equal to the shape function $^a h_{3,u}(\xi_1, \xi_2)$ in global coordinates. Here, the first derivative is equal to zero at $x \in \{0, L/2\}$. If we normalize the peak of the first mode to 1, the FEM approximation can be extracted.

$$\begin{aligned}
 \underline{n = 2}: \quad & h_{3,u}(x, y = L/2) = -120x^3 + 48x^2 \quad x \in [0, L/2] \\
 \underline{n = 4}: \quad & h_{3,u}(x, y = L/2) = \begin{cases} \frac{1}{\sqrt{2}} (-1024x^3 + 192x^2), & x \in [0, L/4] \\ \left(1 - \frac{1}{\sqrt{2}}\right) \left(-1024x^3 + 576x^2 - 96x + 6 + \sqrt{2}\right), & x \in]L/4, L/2] \end{cases} \\
 & \vdots \\
 & n = 8, 16, 32, \dots
 \end{aligned}$$

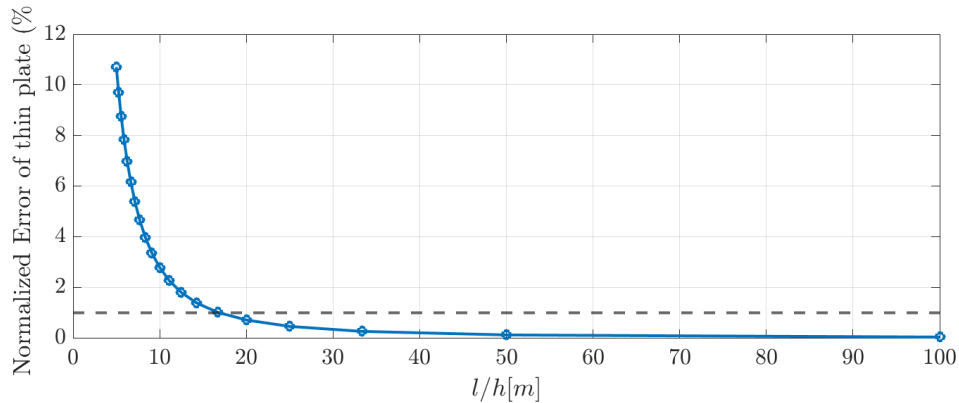
where as $n \rightarrow \infty$, the piecewise cubic polynomial progressively fits more accurately the sine curve of the first mode shape, and the relative error of the shape decreases. The oscillations induced by the properties of the cubic polynomials are progressively diluted as the number of elements increases.

3.3 Study of the ℓ/h ratio

Both ACM and CR element methods are fundamentally grounded in classical thin-plate (Kirchhoff-Love, Eq. 23) theory, which assumes negligible transverse shear deformation and rotary inertia. As a result, ACM and CR methods are most accurate for thin plates where these effects are negligible. For thicker plates or when higher-order deformation modes become significant, the assumptions inherent in thin-plate theory lead to inaccurate frequency predictions and mode shapes. Consequently, while ACM and CR methods are efficient and accurate for thin plate problems, their predictive capabilities degrade in thicker geometries, requiring more refined models such as those based on Mindlin-Reissner theory (Eq. 24) to capture the full behavior.



(a) First natural frequency as estimated by thin and thick plate theory. Length (l) is held constant at 0.5 meters.



(b) Error of thin plate natural frequency with relation to thick plate. The grey line indicates a 1% error cutoff.

Figure 5: Visualization of the variations in predictions of the first natural plate frequency using thick and thin plate theory over a range of length to thickness ratios.

Figure 5 shows the discrepancy between the thin plate models with the thick plate Mindlin-Reissner theory for varying length to height ratios. While the models give very similar values at low plate thicknesses h (or high l/h ratios), the models rapidly diverge beginning around the $l/h = 20$ point. If the goal is to maintain an error of less than 1%, the l/h ratio must remain above ≈ 16.67 .

4 Conclusion

Finite element method applied to a thin plate using the Kirchhoff-Love theory (Equation 23) was successfully implemented using both ACM and CR elements. The influence of element type, number of elements, and plate slenderness ratio (l/h) on the accuracy of first natural frequency predictions was evaluated and compared against analytical models.

The CR element demonstrated superior convergence behavior, achieving error less than 0.02% with as few as 4×4 elements, while ACM elements required finer meshes to approach similar accuracy. This performance is attributed to the CR element's conformity to C^1 continuity and its higher order polynomial basis, which increases stiffness and results in a slight overestimation of natural frequencies. In contrast, ACM elements underestimate the frequency due to non-conforming shape functions and less constrained DOF. Solving the eigenvalue problem takes more time using CR elements, but only minimally more, meaning it still creates a significant improvement over ACM elements.

Comparison across varying l/h ratios illustrates that both ACM and CR methods remain accurate only within the bounds of thin plate assumptions. Since the Kirchhoff-Love method is based on thin plane theory, it handles well plates with high aspect ratios l/h . However since transverse shear and rotary inertia are neglected, when the aspect ratio of the plate becomes too small, the error on the natural frequency increases. This error is limited to 1% when the aspect ratio is decreased to $l/h \approx 17$, meaning that the Kirchhoff-Love method is appropriate for all plate cases with an aspect ratio above $l/h = 17$. When the aspect ratio becomes too large for the Kirchhoff-Love method, the Reissner-Mindlin method (Eq. 24) may be used to account for transverse shear and inertia.

While this study investigated the capabilities of CR and AMC elements, these are only two of over 88 element types which exist within plate bending theory for diverse applications and complexity. This study analyses only a very small portion of possible element types and different results could be achieved if the complexity and/or DOF of the elements are varied.

5 References

1. Stefano Burzio, *Dynamic finite element analysis of structures* [ME-473]. EPFL (2025).